

POLITECNICO
MILANO 1863

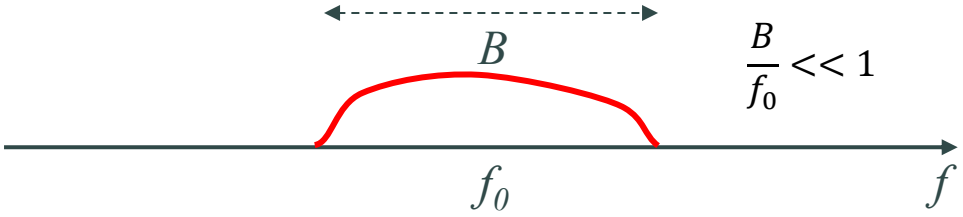
DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA

APPLIED ELECTROMAGNETICS @POLIMI

RF Systems

RF Signals & Envelope
Frequency Conversion
Gain

RF Signal



f: frequency (Hz)
$\omega = 2\pi f$ angular frequency or pulsation (rad/s)

Analytic formulation of a real RF voltage (amplitude and phase modulated):

$$s_{RF}(t) = V_M(t)\cos\big(\omega_0 t + \Phi(t)\big) = V_I(t)\cos(\omega_0 t) - V_Q(t)\sin(\omega_0 t)$$

Pure amplitude modulation: $V_M(t)\cos\omega_0 t$

Pure phase modulation: $\cos(\omega_0 t + \Phi(t))$

$V_M(t)$, $\Phi(t)$: Magnitude and phase (instantaneous)
$V_I(t)$, $V_Q(t)$: In phase and Quadrature signal components
Information associated to RF Signal is carried by these signals only

$\cos(\omega_0 t)$, $\sin(\omega_0 t)$: carrier signal, not conveying any information

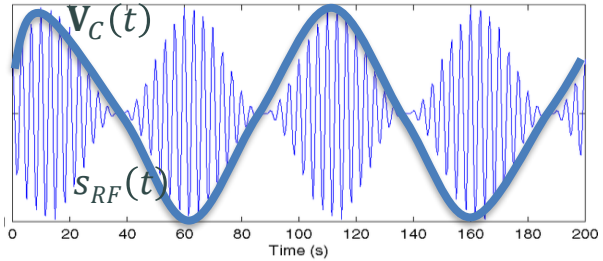
Complex envelope representation

Phasor notation: when a signal is at ω_0, $v(t)=A\cos(\omega_0 t+\phi)=\mathrm{Re}\{A\cdot e^{j\phi}\cdot e^{j\omega_0 t}\}$ → $\mathbf{V}=A\cdot e^{j\phi}$ is the phasor representation of v(t)

RF signal:

$$s_{RF}(t)=V_M(t)\cos\big(\omega_0 t+\Phi(t)\big)=V_I(t)\cos(\omega_0 t)-V_Q(t)\sin(\omega_0 t)$$
$$V_I(t)=V_M(t)\cos\big(\Phi(t)\big),\qquad V_Q(t)=V_M(t)\sin\big(\Phi(t)\big)$$

$$\cos\theta\,\cos\varphi=\frac{\cos(\theta-\varphi)+\cos(\theta+\varphi)}{2}$$



Use same concept on a RF signal:

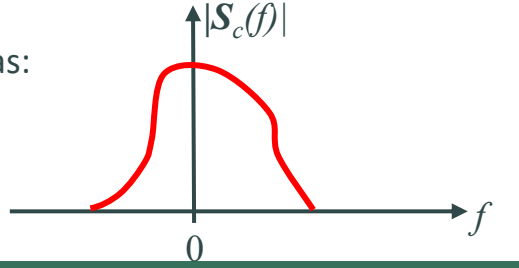
$$s_{RF}(t)=\mathrm{Re}\{V_M(t)e^{j\Phi(t)}e^{j\omega_0 t}\}=\mathrm{Re}\{\mathbf{V}_C(t)e^{j\omega_0 t}\}$$

Complex envelope of $s_{RF}(t)$: $\mathbf{V}_C(t)=V_M(t)e^{j\Phi(t)}=V_I(t)+jV_Q(t)$

The spectrum of $\mathbf{V}_c(t)$ is around f=0 and contains the same information

The spectrum of $\boldsymbol{V}_c(t)$ (namely $\boldsymbol{S}_c(f)$) is related to the spectrum of $s_{RF}(t)$ (namely $S_{RF}(f)$) as:

$$S_{RF}(f)=F\left\{Re\{V_C(t)e^{j\omega_0 t}\}\right\}=\frac{1}{2}S_C(f-f_0)+\frac{1}{2}S_C^*(-f-f_0)$$



To verify spectrum of RF signal from envelope

Definitions:

$$\mathbf{V}_C(t) = V_M(t)e^{j\Phi(t)} = V_I(t) + jV_Q(t)$$
$$S_C(f) = F\{V_C(t)\} = S_I(f) + jS_Q(f)$$

Things to remember from Fourier transform:

$F\{x(t)y(t)\} = X(f) * Y(f)$ $\quad [* \, means \, convolution]$
$Y(f) = F\{y(t)\} \rightarrow F\{y^*(t)\} = Y^*(-f)$
$X(f) = F\{x(t)\}$: if x(t) is real, X(f) can be complex but $X(-f) = X^*(-f)$

First way:

$$s_{RF}(t) = V_I(t)\cos(\omega_0 t) - V_Q(t)\sin(\omega_0 t)$$

$$S_{RF}(f) = F\{V_I(t)\cos(\omega_0 t)\} - F\{V_Q(t)\sin(\omega_0 t)\}$$
$$= S_I(f) * \frac{\delta(f-f_0)+\delta(f+f_0)}{2} - S_Q(f) * \frac{\delta(f-f_0)-\delta(f+f_0)}{2j}$$
$$= \frac{1}{2}S_I(f-f_0) + \frac{1}{2}S_I(f+f_0) + \frac{j}{2}S_Q(f-f_0) - \frac{j}{2}S_Q(f+f_0)$$
$$= \frac{1}{2}S_C(f-f_0) + \frac{1}{2}S_I^*(-f-f_0) - \frac{j}{2}S_Q^*(-f-f_0)$$
$$= \frac{1}{2}S_C(f-f_0) + \frac{1}{2}\left(S_I\ (-f-f_0) + \frac{j}{2}S_Q\ (-f-f_0)\right)^*$$

Second way:

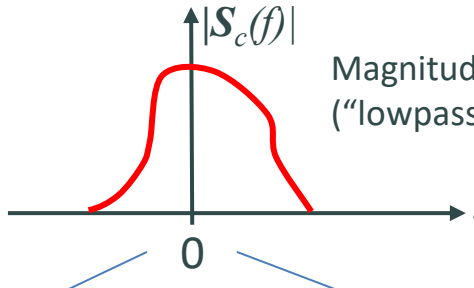
$$s_{RF}(t) = \mathrm{Re}\{V_M(t)e^{j\Phi(t)}e^{j\omega_0 t}\} = \mathrm{Re}\{\mathbf{V}_C(t)e^{j\omega_0 t}\}$$

$$S_{RF}(f) = F\left\{Re\{V_C(t)e^{j\omega_0 t}\}\right\}$$
$$= \frac{1}{2}F\{V_C(t)e^{j\omega_0 t} + V_C^*(t)e^{-j\omega_0 t}\}$$
$$= \frac{1}{2}S_C(f) * \delta(f-f_0) + \frac{1}{2}S_C^*(-f) * \delta(f+f_0)$$

$$S_{RF}(f) = \frac{1}{2}S_C(f-f_0) + \frac{1}{2}S_C^*(-f-f_0)$$

Spectrum

$$S_{RF}(f) = F\left\{Re\left\{V_C(t)e^{j\omega_0 t}\right\}\right\} = \frac{1}{2}S_C(f - f_0) + \frac{1}{2}S_C^*(-f - f_0)$$



Magnitude of complex envelope spectrum ("lowpass")

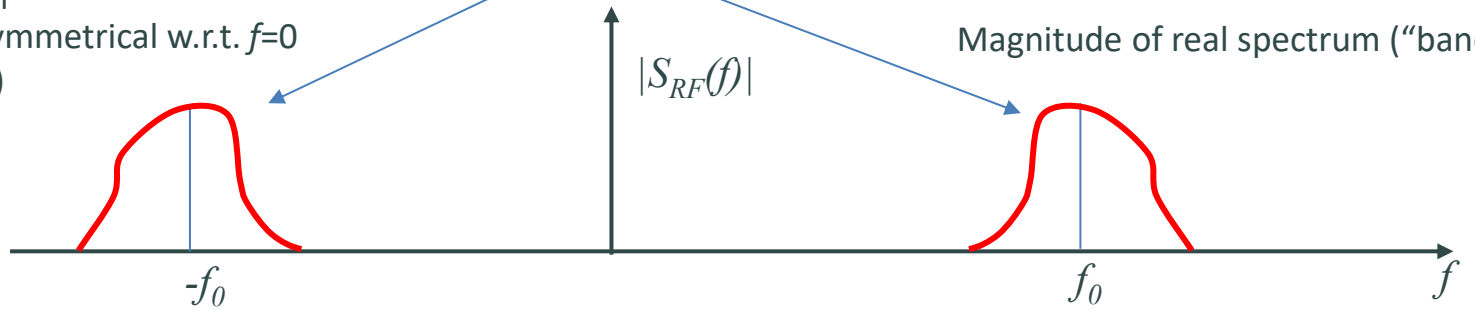
For a real signal (such as $s_{RF}(t)$)

→ Magnitude is symmetrical w.r.t. f=0

$|F(-f)|=|F(f)|$

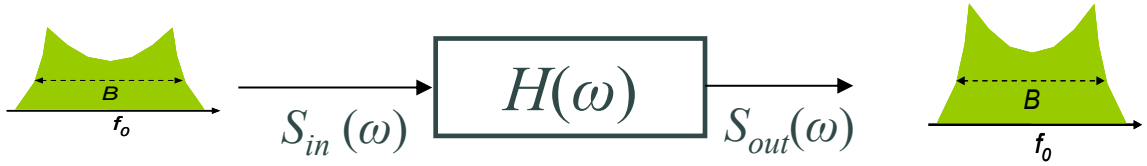
→ Phase is anti-symmetrical w.r.t. f=0

$\angle F(-f)=-\angle F(f)$



Magnitude of real spectrum ("bandpass")

RF Signal across a linear 2-port: no distortion condition



Given the input RF signal in the frequency domain (S_{in}), the output is

$$S_{out}(\omega) = H(\omega) \cdot S_{in}(\omega)$$

To avoid a degradation of the output signal (=loss of information), the transfer function must satisfy:

$|H(\omega)| = constant$ → Flat magnitude

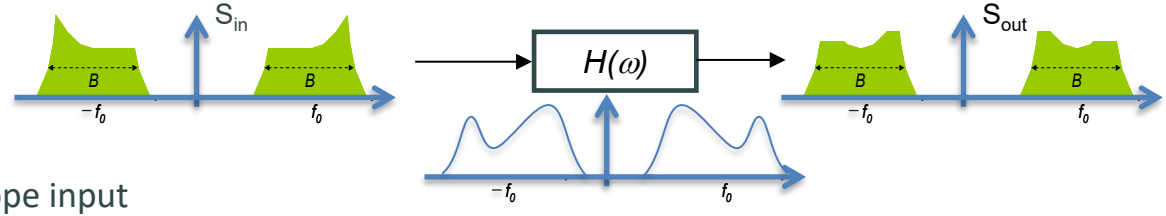
$\angle\big(H(\omega)\big) \propto \omega + constant$ → Linear phase response

These conditions are requested in the signal band only (bandpass characteristic)

Most components only "try" to approach the "no distortion condition"

Baseband equivalent of a 2-port

$$S_{out}(\omega)=H(\omega)\cdot S_{in}(\omega)$$



Baseband frequency: $\Omega=\omega-\omega_0$

$C_{in}(\Omega)$: spectrum of the complex envelope input

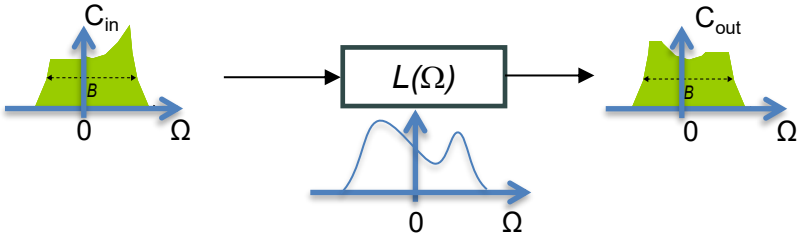
$C_{out}(\Omega)$: spectrum of the complex envelope output

$$C_{out}(\omega-\omega_0)+C^*_{out}(\omega+\omega_0)=H(\omega)\cdot\left(C_{in}(\omega-\omega_0)+C^*_{in}(\omega+\omega_0)\right)$$

$$C_{out}(\omega-\omega_0)=H(\omega)\cdot C_{in}(\omega-\omega_0)$$

$$C_{out}(\Omega)=H(\Omega+\omega_0)\cdot C_{in}(\Omega)$$

$$C_{out}(\Omega)=L(\Omega)\cdot C_{in}(\Omega)$$

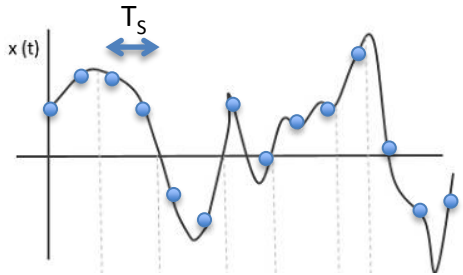


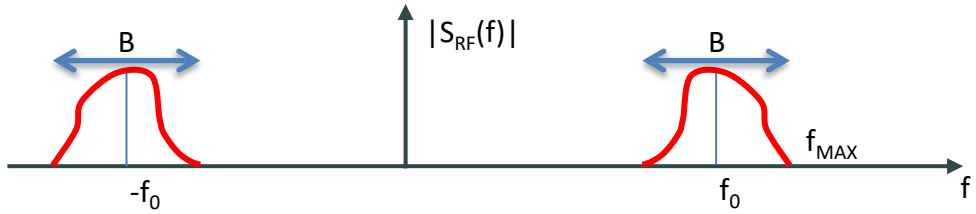
$L(\Omega)$ is a lowpass function (centered around zero frequency)

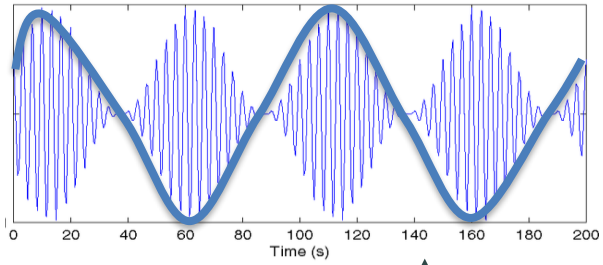
Complex envelope equivalence is meaningful

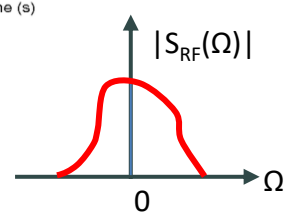
only when the transient response is exhausted (steady-state condition)

Advantage of using the complex envelope representation









- Minimum sampling step for a signal (Nyquist theorem):

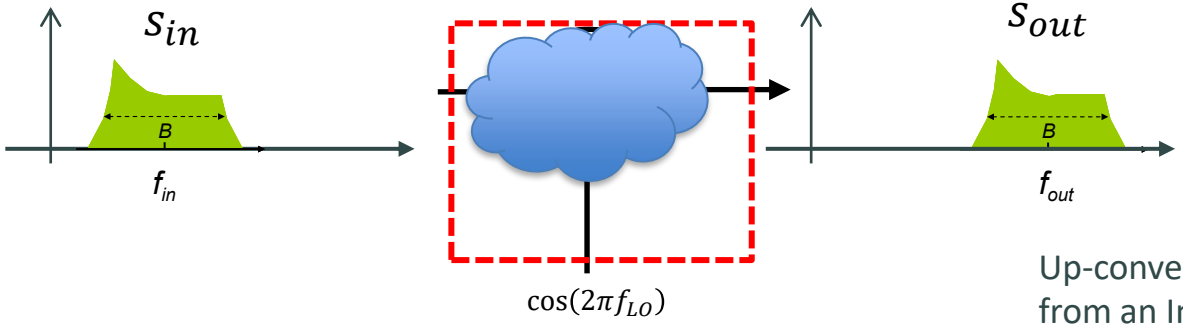
$$T_S < \frac{1}{2fMax}$$

- On the RF signal: f_0=1GHz, B=10MHz

  Ts <0.5ns; → on its cplx envelope Ts<100ns

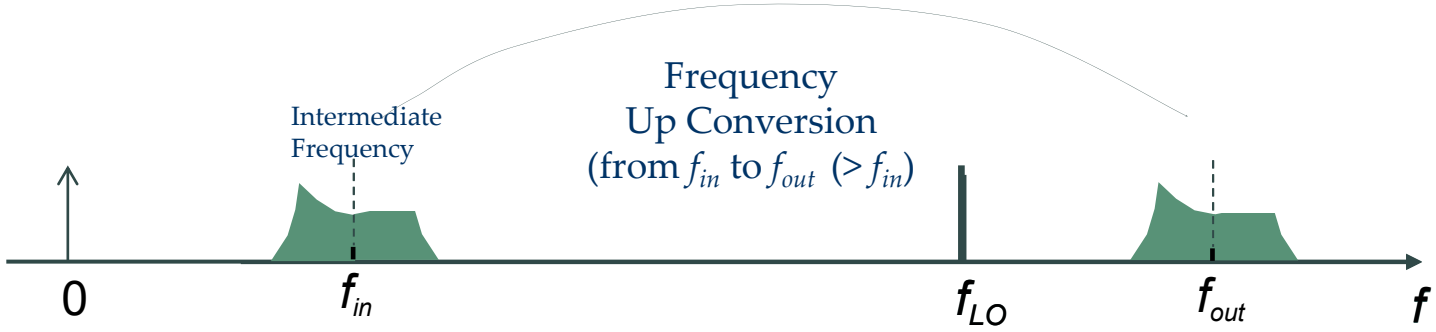
- In a simulation software (i.e. CAD simulators):
  → a huge memory/computation time to simulate in the bandpass domain
  → much more affordable in complex envelope, without accuracy loss

Frequency Conversion

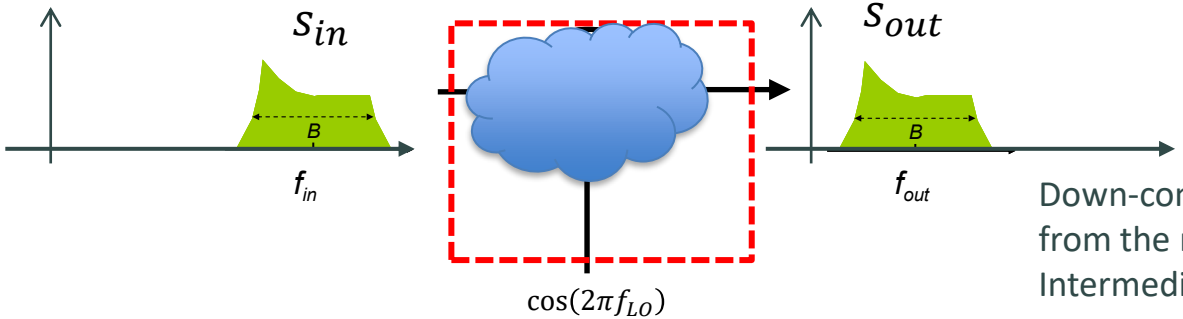


$$f_{out} = |f_{in} \pm f_{LO}|$$

Up-conversion is usually done in the transmitter, from an Intermediate Frequency (IF) to the required transmission frequency



Frequency Conversion



$$f_{out} = |f_{in} \pm f_{LO}|$$

Down-conversion is usually done in the receiver, from the required received frequency to an Intermediate Frequency (IF)

Frequency Down Conversion (from f_{in} to f_{out} ($< f_{in}$)

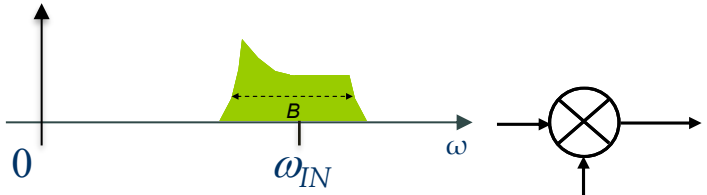


Frequency conversion between two RF signals: how?

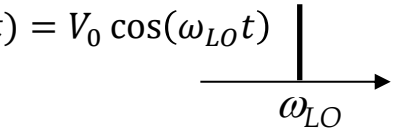
By multiplying the input signal by a sinusoidal tone (local oscillator) through a MIXER

$$v_{OUT}(t) = v_{IN}(t)\, v_{LO}(t)$$

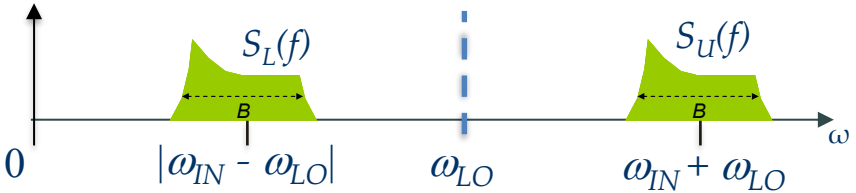
$$S_{OUT}(f) = S_{IN}(f) * \frac{\delta(f-f_0)+\delta(f+f_0)}{2} = \frac{V_0}{2} S_{IN}(f-f_{LO}) + \frac{V_0}{2} S_{IN}(f+f_{LO})$$



$$v_{IN}(t) = V_M(t)\cos\left(\omega_{IN} t + \phi(t)\right)$$



$$v_{LO}(t) = V_0 \cos(\omega_{LO} t)$$



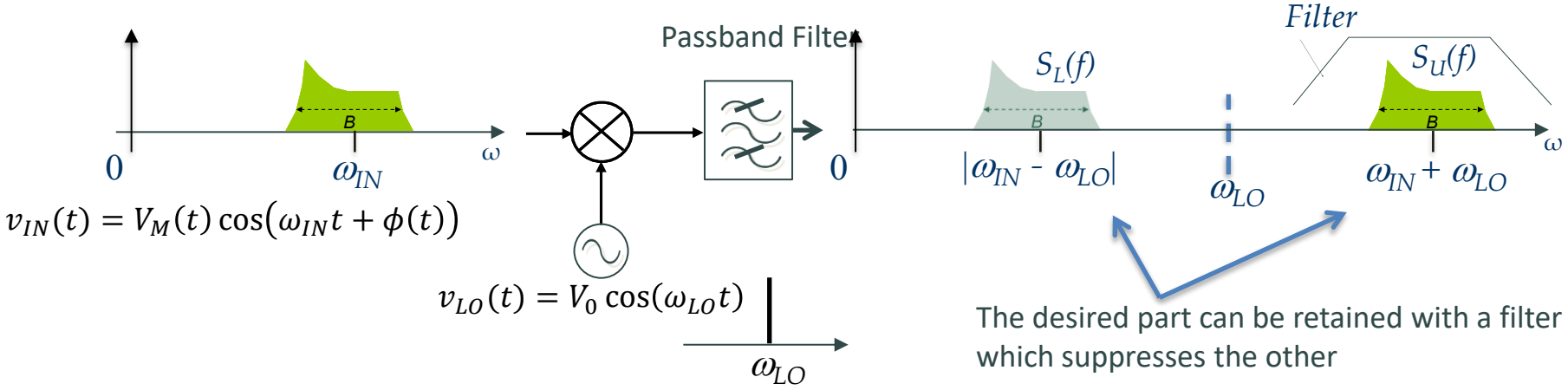
Frequency conversion: how?

By multiplying the input signal by a sinusoidal tone (local oscillator) through a MIXER

$$v_{OUT}(t) = v_{IN}(t)\, v_{LO}(t)$$

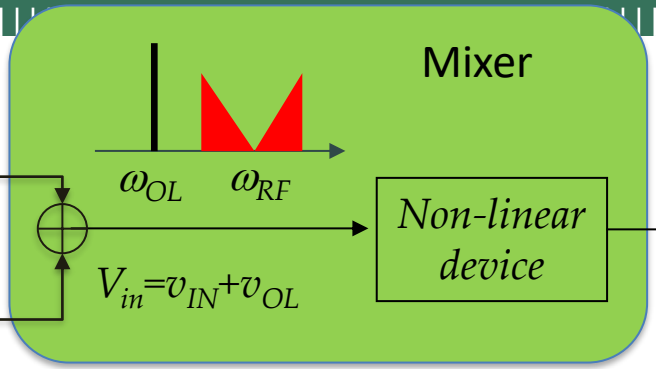
$$S_{OUT}(f) = \frac{V_0}{2} S_{IN}(f - f_{LO}) + \frac{V_0}{2} S_{IN}(f + f_{LO})$$

We are usually not allowed to occupy two bands!
→ Suppress the undesired sideband via filtering



The desired part can be retained with a filter which suppresses the other

Alternative mixer: exploiting the non-linearity of a device



$v_{IN}(t) = V_M(t)\cos(\omega_{IN}t + \phi(t))$

$v_{OL}(t) = V_0\cos(\omega_{OL}t)$

$V_{out} = a_1 V_{in} + a_2 V_{in}^2 + \ldots$

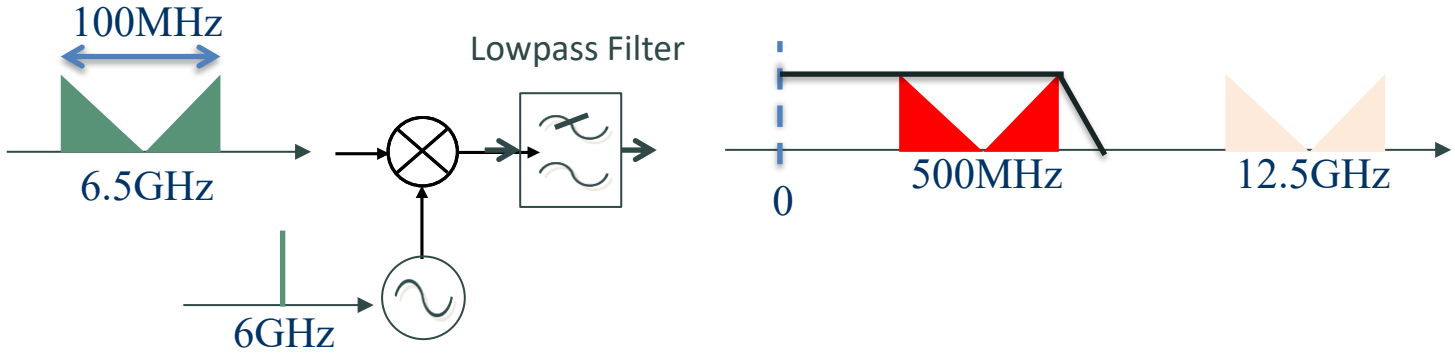
$$V_{out} = \frac{a_2}{2}(V_0 + V_M) + a_1(V_0\cos(\omega_{OL}t) + V_M\cos(\omega_{RF}t)) +$$
$$+\frac{a_2}{2}(V_0^2\cos(2\omega_{OL}t) + V_M^2\cos(2\omega_{RF}t)) + a_2(2V_0V_M\cos(\omega_{OL}t)\cdot\cos(\omega_{RF}t)) + \ldots$$

At the output we select the desired component with a Filter

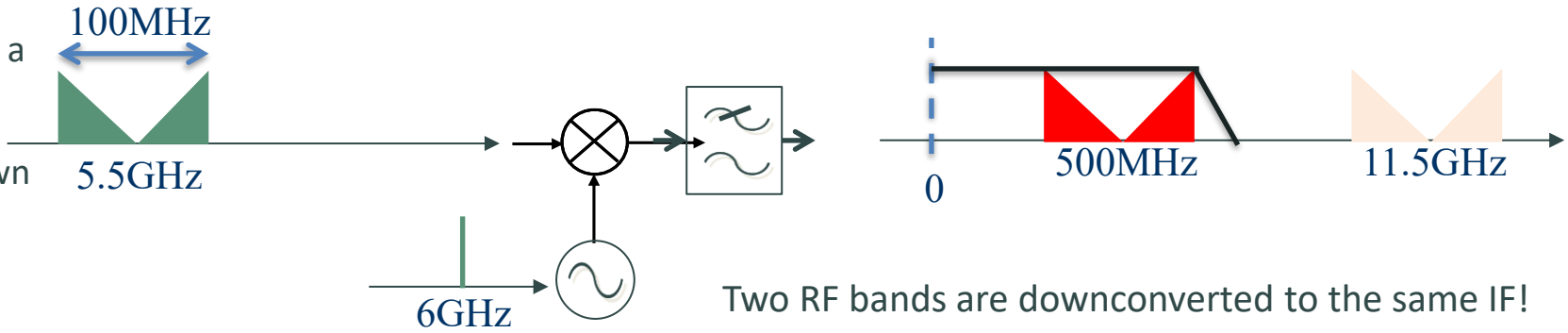


Image Band in Downconversion

(previous example: bring a desired RF signal from 6.5GHz down to 500MHz)

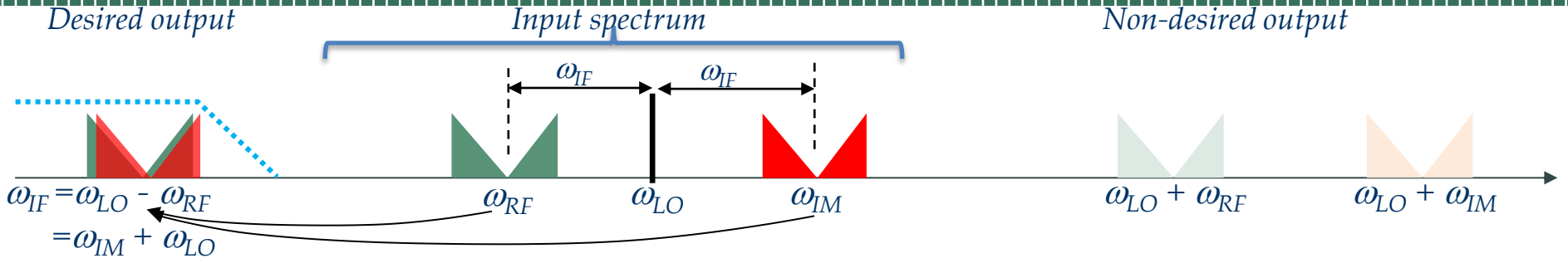


Now, bring a desired RF signal from 5.5GHz down to 500MHz



Two RF bands are downconverted to the same IF!

Image Band in Downconversion



The image band is centered at $2\omega_{IF}$ from ω_{RF} and has the same band as the useful one

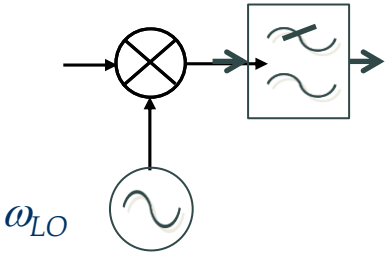
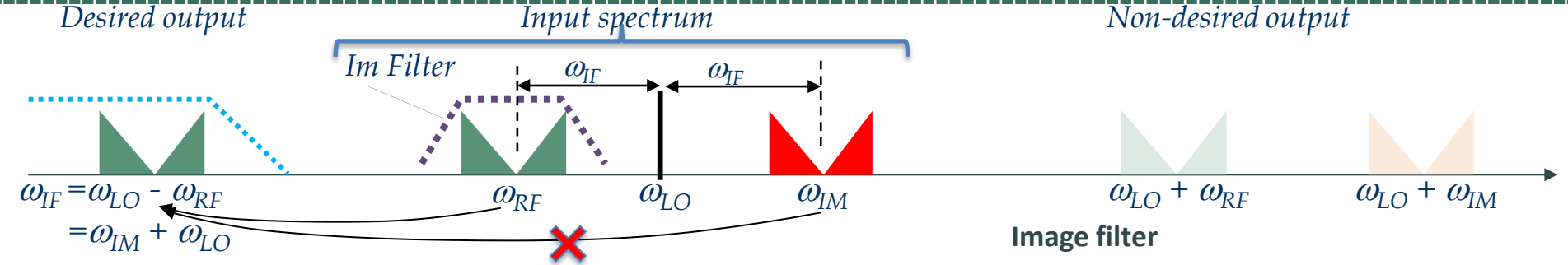
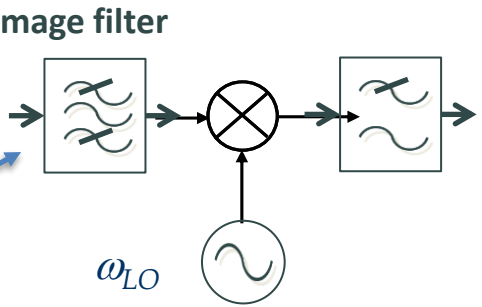


Image Band in Downconversion



The image band is centered at $2\omega_{IF}$ from ω_{RF} and has the same band as the useful one

→ Add a bandpass filter at ω_{RF} to suppress signals in the image band!



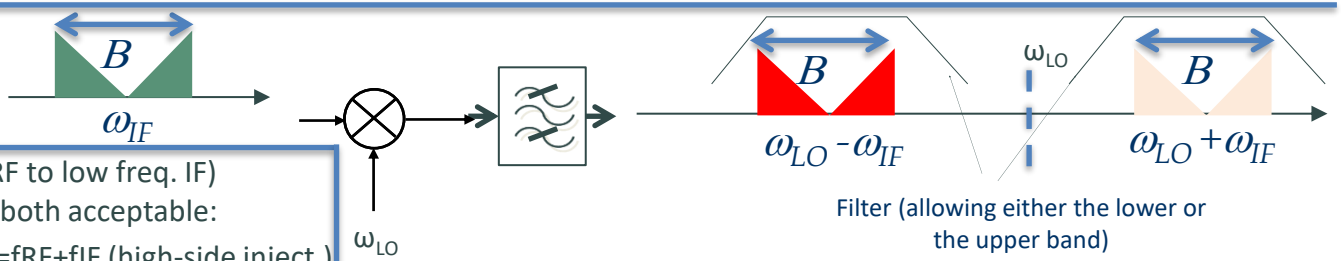
Practical note: in practice the filter after the mixer in downconversion is often created by the other components in the chain, which are designed to work at ω_{IF} and thus automatically reject high frequencies (ω_{LO} and above)
→ In our exercises we will not indicate the LPF, but we will act as if it is there, unless explicitly stated otherwise.

Take aways

Multiplication of a real signal around ω_{IN} by a carrier ω_{LO} creates two equal bands around:

"sum" frequency ($|\omega_{IN}+\omega_{LO}|$) **"difference" frequency** ($|\omega_{IN}-\omega_{LO}|$) → One needs to be suppressed (sometimes implicitly done by next components)

Up-conversion
(from low freq. IF to high freq. RF)



Filter (allowing either the lower or the upper band)

Down-conversion (from high freq. RF to low freq. IF)

- Two LO frequencies can be chosen, both acceptable:
  fLO=fRF-fIF (low-side inject.) OR fLO=fRF+fIF (high-side inject.)
- Whatever choice of fLO>0, an image band will be downconverted to the same IF:

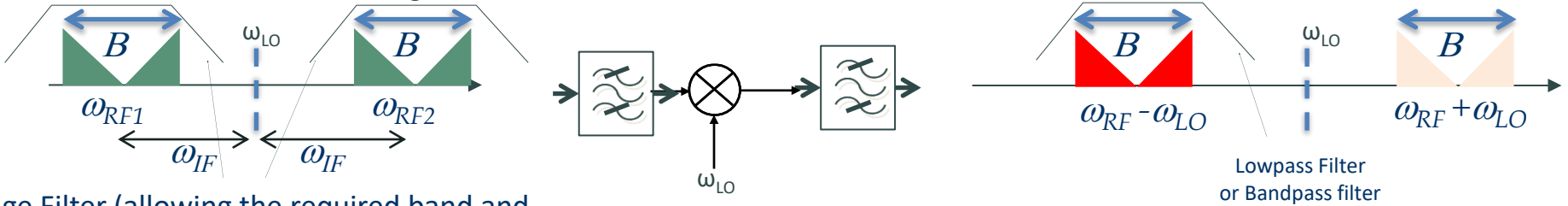
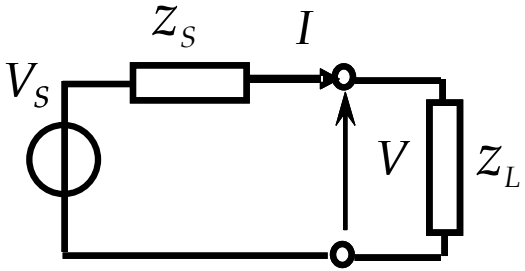


Image Filter (allowing the required band and rejecting the other («image»))

Lowpass Filter
or Bandpass filter

«image band» is at $2\omega_{IF}$ from the desired band (on the other side of ω_{LO})

Available power from a source and conjugate matching



The Available power P_{av} of a source is the maximum power the source can deliver to a load.
This happens when the load impedance is equal to the conjugate of the source impedance (**conjugate matching**)

P_{av} is a property of the source:

$$P_{av} = \frac{1}{8}\frac{|V_S|^2}{\mathrm{Re}(Z_S)}$$

A load Z_L can absorb at most P_{av}: this happens if

$$Z_L = Z_S^* \quad \Rightarrow \quad P_L = P_{av}$$

To verify: write V as function of V_S, $Z_S=R_S+jX_S$, $Z_L=R_L+jX_L$
Write real power dissipated by the load $P_L=\mathrm{Re}(V I^*)/2$
Compute derivative of P_L w.r.t. X_L and equate to 0 → $X_L=-X_S$
Replace $X_L=-X_S$ in P_L and then compute derivative w.r.t. R_L
Set it to 0 → $R_L=R_S$
Thus conjugate matching is the condition for maximum power transfer and the obtained P_L (replacing also $R_L=R_S$) is the available power from the generator.

Power Transfer from Source to Load (e.g. Amplifier)



Problem: there are several ways to define the "gain"!

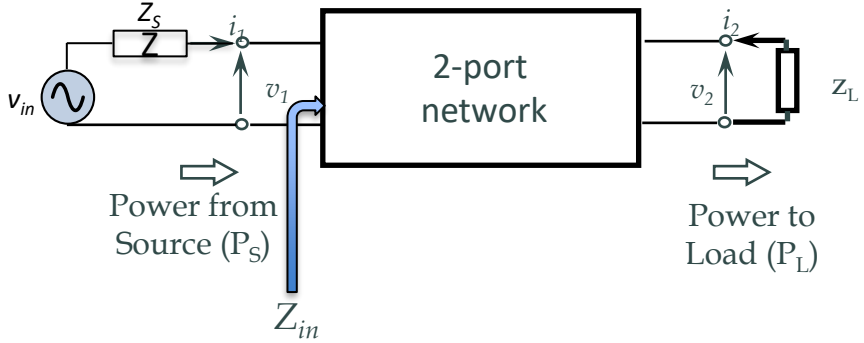
(Sinusoidal steady state regime is assumed)

Definition of Power Gain G_P (also known as "Operating Power Gain")

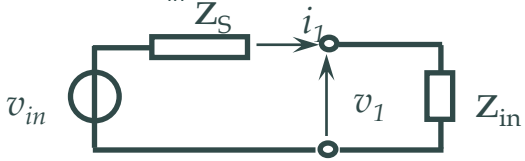
$$P_S = P_{in} = \frac{1}{2}\mathrm{Re}(v_1 \cdot i_1^*) = \text{Power Entering the 2-port}$$

$$P_L = P_{out} = \text{Power delivered to Load} = -\frac{1}{2}\mathrm{Re}(v_2 \cdot i_2^*)$$

$$Gain = G_P = \frac{P_{out}}{P_{in}}$$



BUT: P_{in} may be very small, making G_p unreasonable large. In fact:



$$v_1 = v_{in}\frac{Z_{in}}{Z_{in}+Z_S} \quad \Rightarrow \quad v_1 \text{ goes to 0 for } Z_{in} << Z_S$$

$$i_1 = \frac{v_{in}}{Z_{in}+Z_S} \quad \Rightarrow \quad i_1 \text{ goes to 0 for } Z_{in} >> Z_S$$

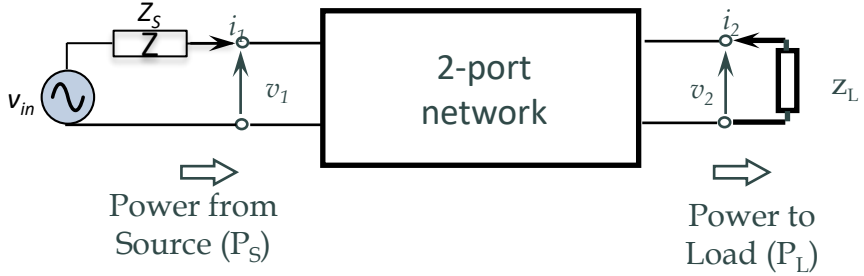
G_p tends to infinite when the 2-port presents a very small (or very large) input impedance (with respect to Z_s)!

Note: A lossless 2-port network (i.e. made only by ideal capacitors, inductors, reactances, transmission lines) has G_P=1

Definition of Transducer Power Gain G_T

$$P_S = \text{Available Power from Source} = P_{AV,in} = \frac{1}{8}\frac{|v_{in}|^2}{\text{Re}(Z_S)}$$

$$P_L = \text{Power delivered to Load} = P_{out} = \frac{1}{2}\left|\frac{v_2}{Z_L}\right|^2 \text{Re}(Z_L)$$

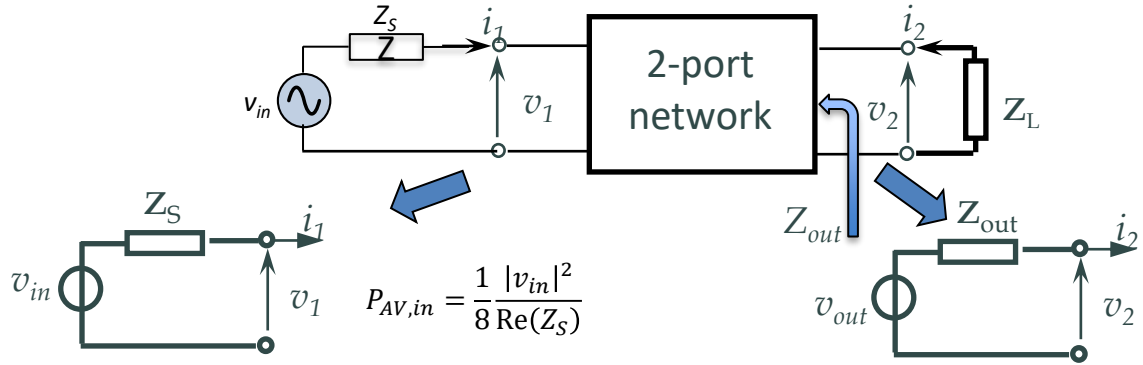


$$Transducer\ Power\ Gain = G_T = \frac{P_{out}}{P_{AV,in}} = 4\frac{\text{Re}(Z_S)\,\text{Re}(Z_L)}{|Z_L|^2}\left|\frac{v_2}{v_{in}}\right|^2$$

G_T represents the power delivered to the load normalized to the maximum power deliverable by the source

G_T best describes the attitude of a device to transfer power to a load (than just a passive matching network connecting the source to the load). Unless otherwise noted, normally Z_S and Z_L = 50Ohm.

Available Power Gain definition: G_A



$$P_{AV,in} = \frac{1}{8}\frac{|v_{in}|^2}{\mathrm{Re}(Z_S)}$$

$$P_{AV,out} = \frac{1}{8}\frac{|v_{out}|^2}{\mathrm{Re}(Z_{out})}$$

$P_{AV,\,in}$=Available power from source

$P_{AV,\,out}$=Available power at output

$$Available\ \text{Power Gain} = G_A = \frac{P_{AV,out}}{P_{AV,in}} = \frac{\mathrm{Re}(Z_S)}{\mathrm{Re}(Z_{out})}\left|\frac{v_{out}}{v_{in}}\right|^2$$

Note: A lossless 2-port network (i.e. made only by ideal capacitors, inductors, reactances, transmission lines) has G_A=1

Features of Power Gains

1. The value of G_T is always smaller than or equal to the value of G_P and G_A (for a given 2-port with assigned Z_L e Z_S)

Transducer Gain G_T ≤ Operating Power Gain G_P ≤ Available Gain G_A

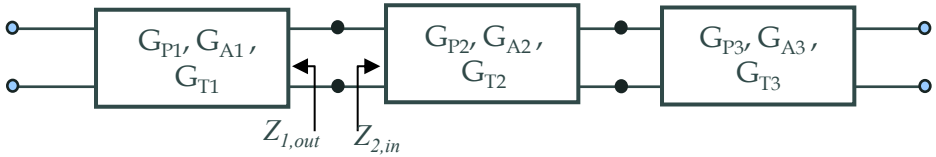
2. If there is conjugate matching at input: $Z_{in} = Z_S^*$ ⟹ $G_T = G_P$
3. If there is conjugate matching at output: $Z_{out} = Z_L^*$ ⟹ $G_T = G_A$



4. If input AND output are conjugately matched, all three gains coincide and G_T is maximized:

If $Z_{in} = Z_S^*$ and $Z_{out} = Z_L^*$ → $G_T = G_P = G_A$

Cascade of 2-ports



The overall G_P and G_A are simply the product of the 2-ports gains:

$$G_P = G_{P1}G_{P2}G_{P3}, \qquad G_A = G_{A1}G_{A2}G_{A3}$$

This does not apply, in general, to the overall transducer G_T

Exception: **if all the 2-ports are conjugately matched**, also the total G_T is the product of the 2-ports G_{Tk}:

$$G_T = G_T G_{T2} G_{T3} \ldots \quad \text{if } Z_{k+1,in} = Z_{k,out}^*$$

General Architecture of a RF Link (unidirectional)

